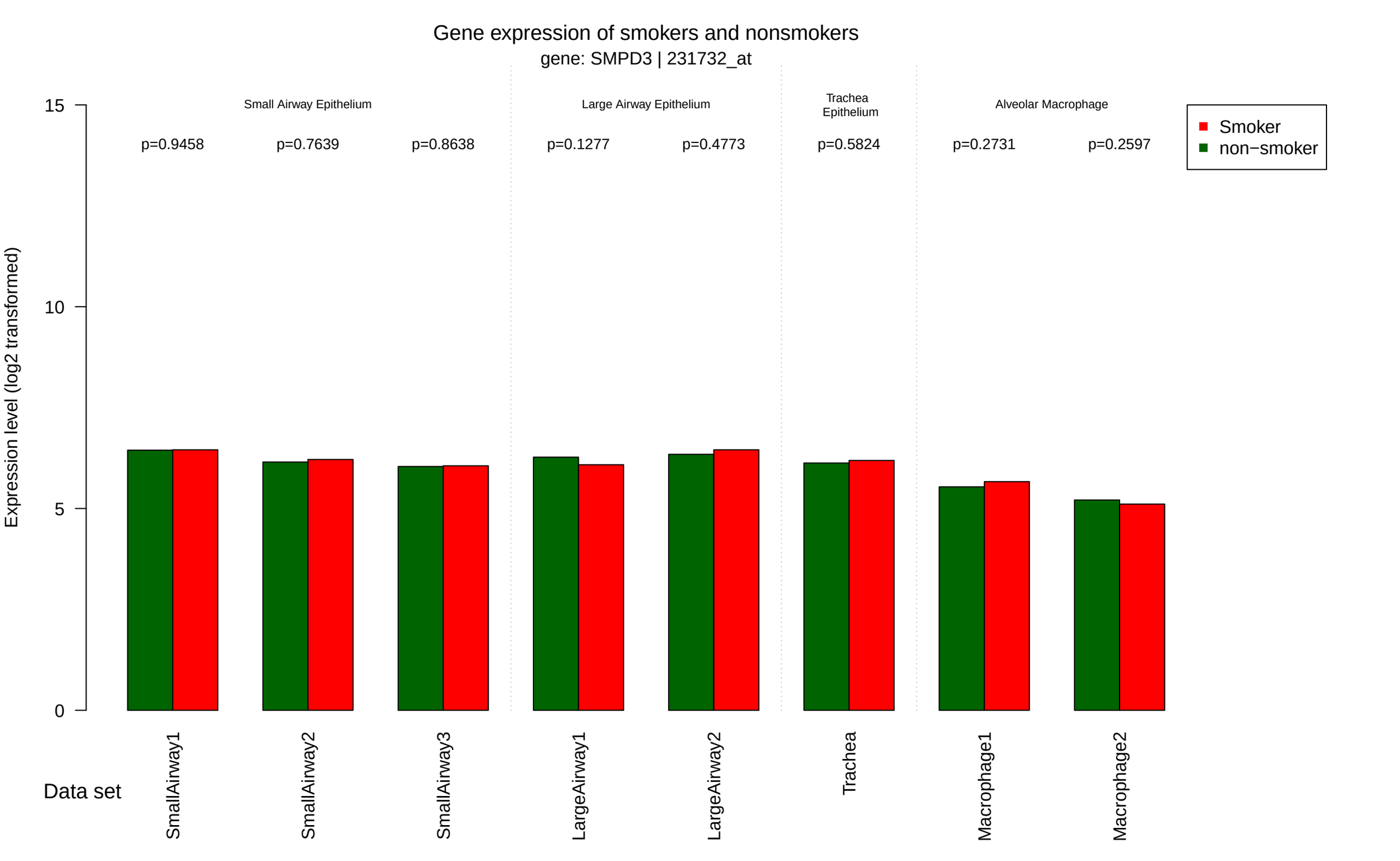
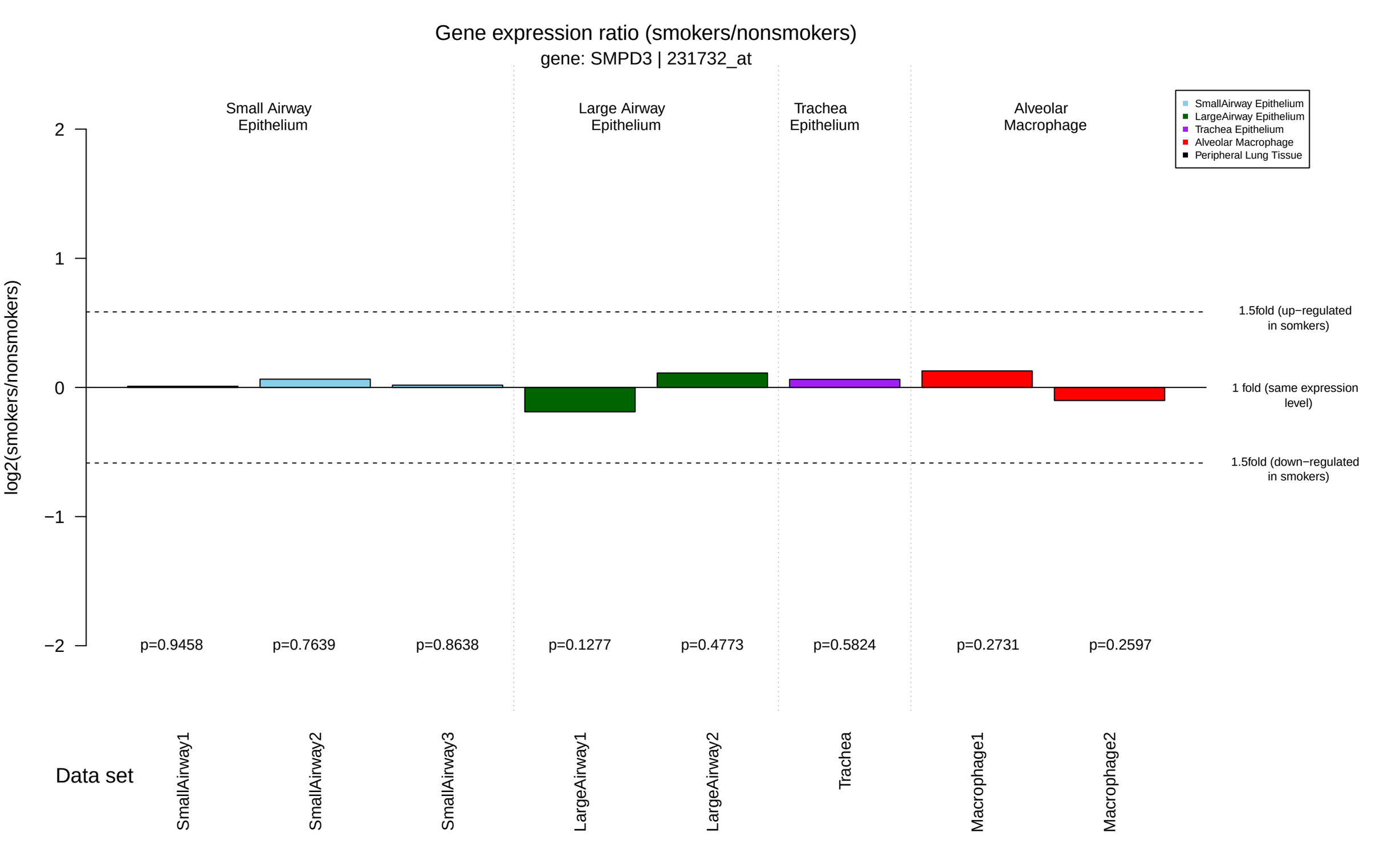
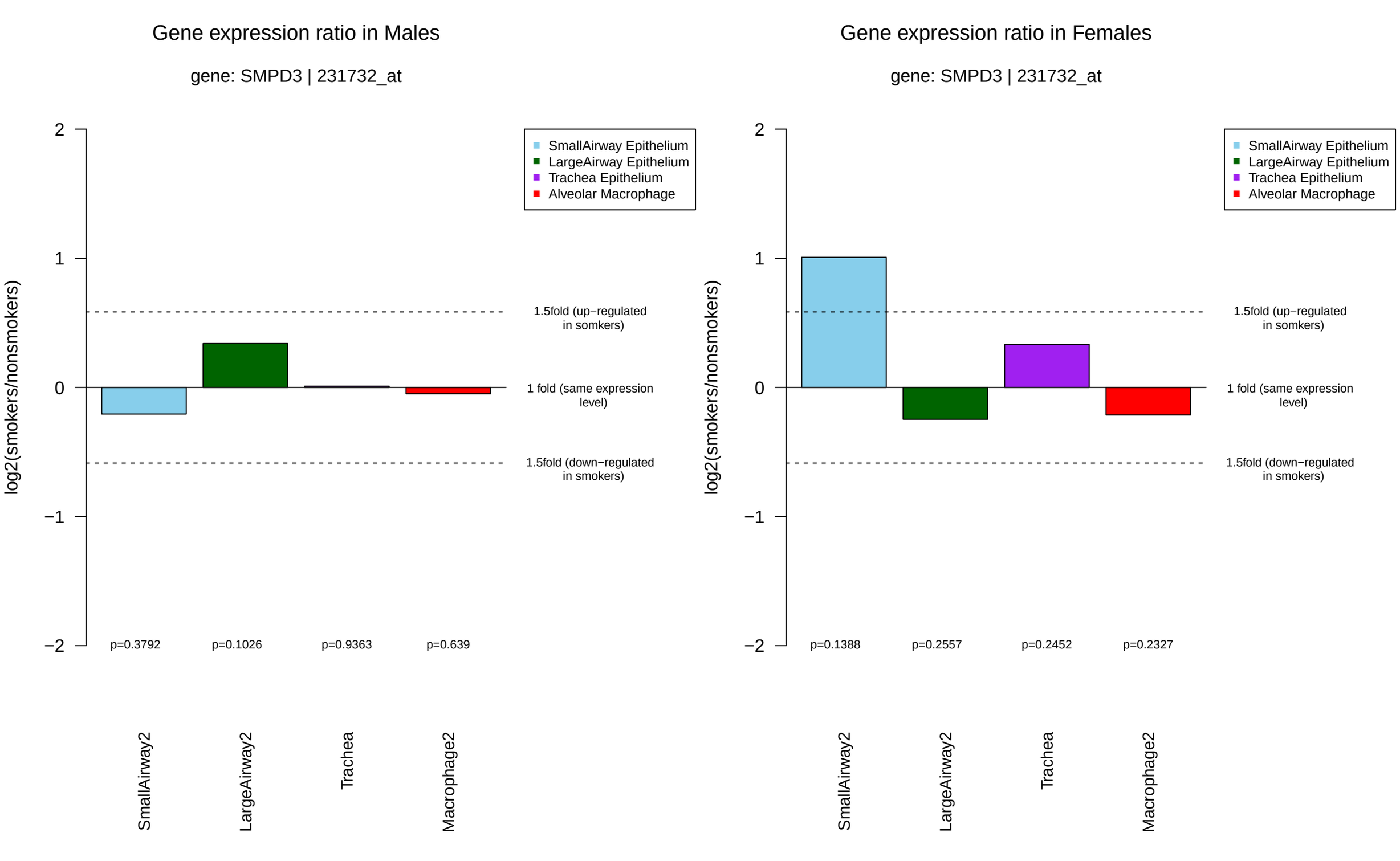
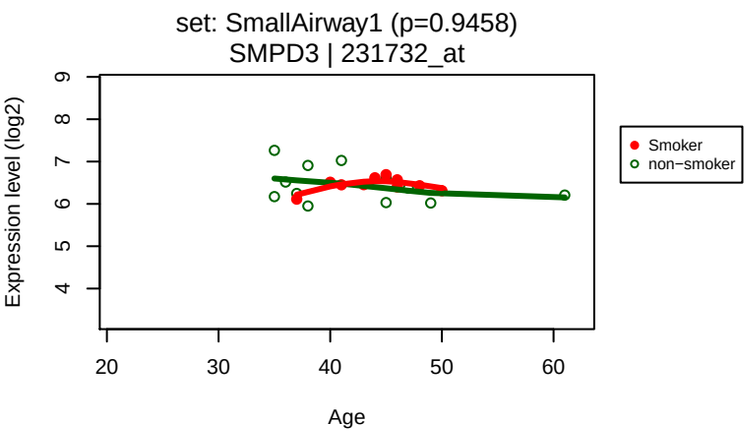
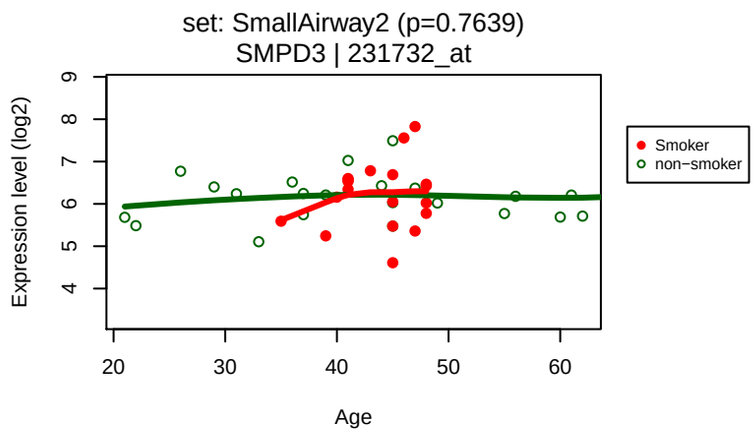
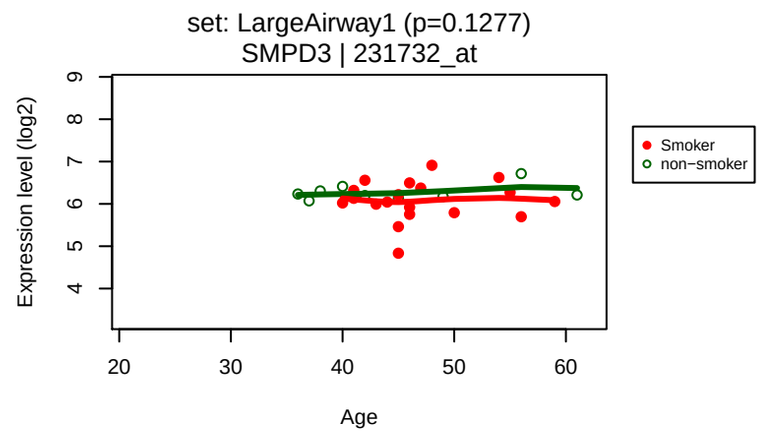

Gene expression of smokers and nonsmokers
gene: SMPD3 | 231732_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9458
p=0.7639
p=0.8638
p=0.1277
p=0.4773
p=0.5824
p=0.2731
p=0.2597
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

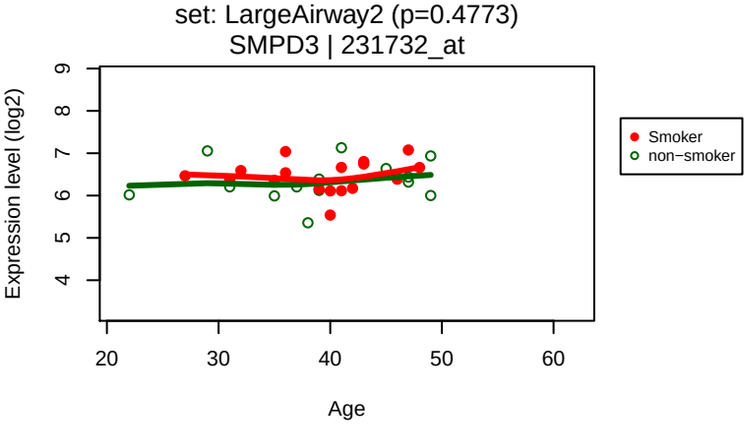

Gene expression ratio (smokers/nonsmokers)
gene: SMPD3 | 231732_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9458
p=0.7639
p=0.8638
p=0.1277
p=0.4773
p=0.5824
p=0.2731
p=0.2597
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

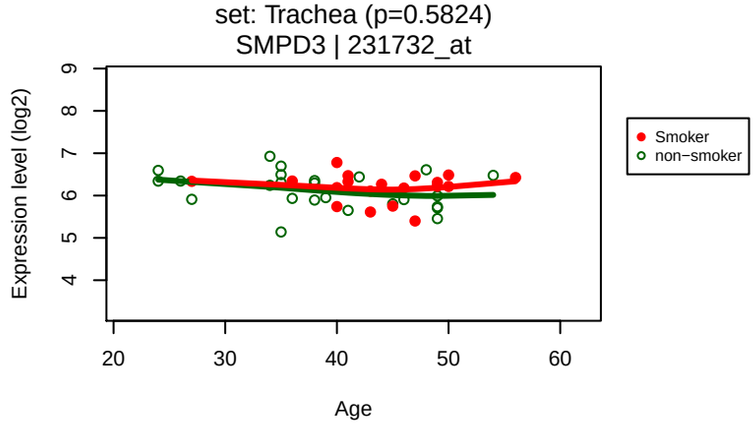

Gene expression ratio in Males
gene: SMPD3 | 231732_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3792
p=0.1026
p=0.9363
p=0.639
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMPD3 | 231732_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1388
p=0.2557
p=0.2452
p=0.2327
SmallAirway2
LargeAirway2
Trachea
Macrophage2

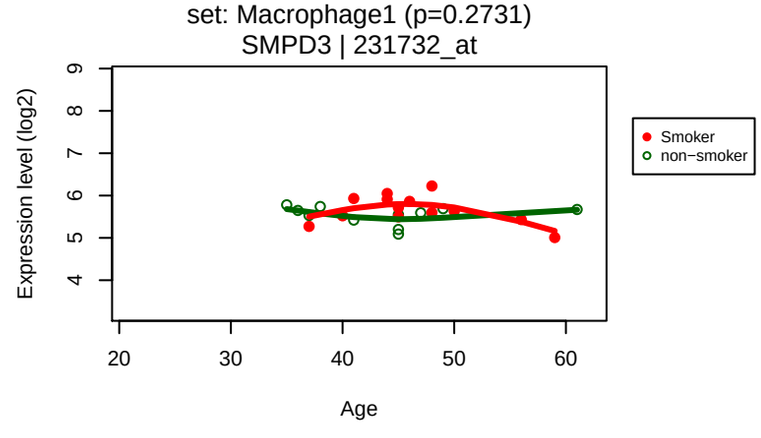

set: SmallAirway1 (p=0.9458)
SMPD3 | 231732_at
Expression level (log2)
Age
Smoker
non−smoker

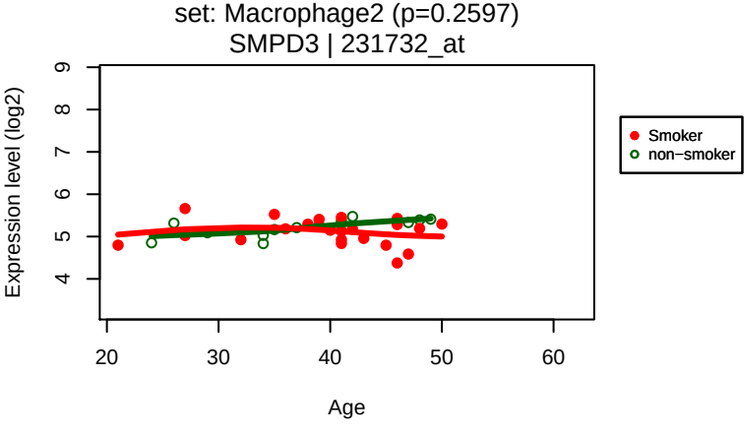

set: SmallAirway2 (p=0.7639)
SMPD3 | 231732_at
Expression level (log2)
Age
Smoker
non−smoker

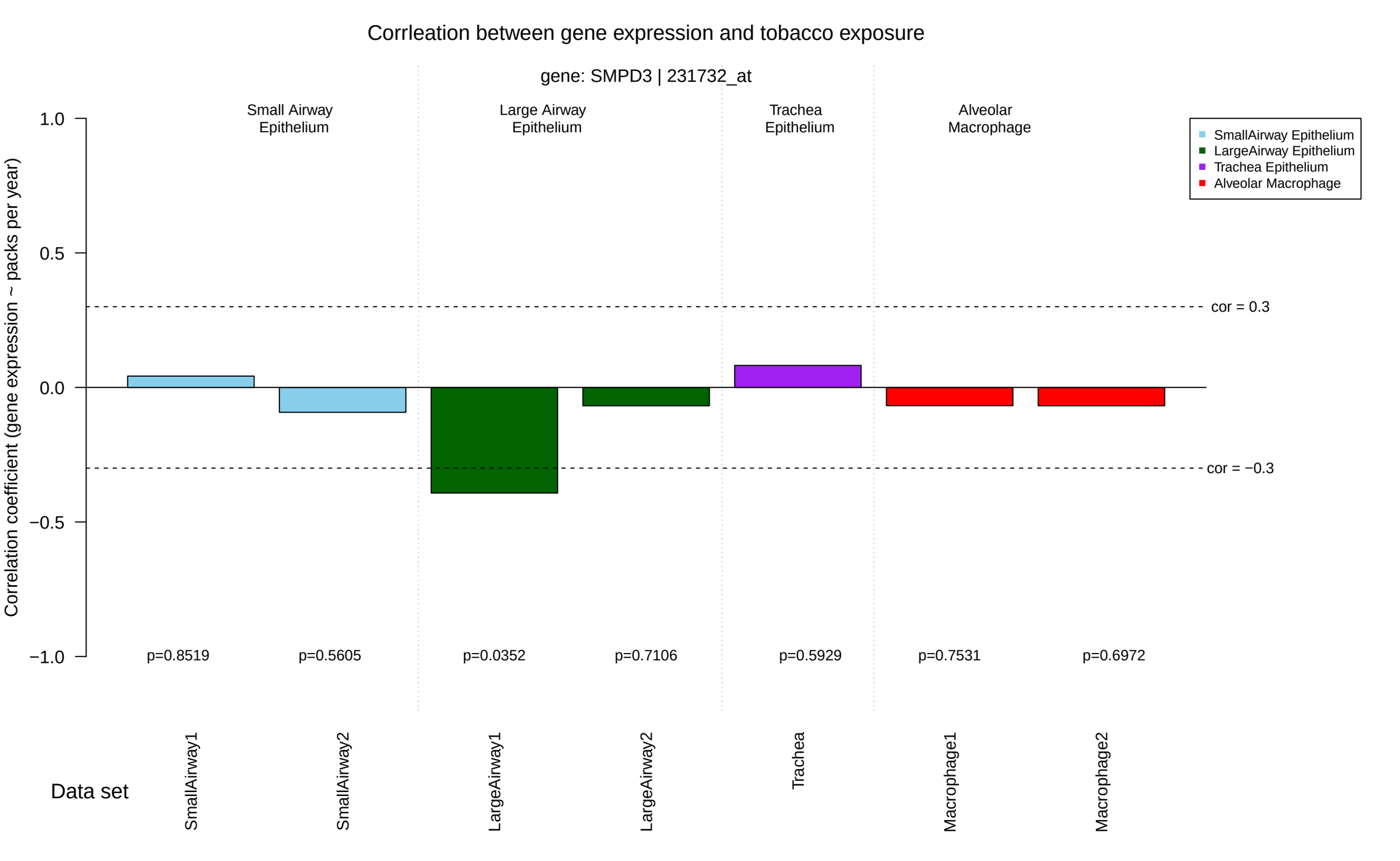

set: LargeAirway1 (p=0.1277)
SMPD3 | 231732_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4773)
SMPD3 | 231732_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5824)
SMPD3 | 231732_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2731)
SMPD3 | 231732_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2597)
SMPD3 | 231732_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMPD3 | 231732_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8519
p=0.5605
p=0.0352
p=0.7106
p=0.5929
p=0.7531
p=0.6972
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2